Venkata Lakshmi Parimala Pasupuleti

+1-703-261-5410 | venkatalakshmiparimala@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti |

AgentProbe: github.com/venkatalakshmiparimala/AgentProbe | Virginia, USA

PROFESSIONAL SUMMARY

Applied AI engineer specializing in **production agentic workflows and LLM evaluation** - built AgentProbe, a production LangGraph system with ReAct/Plan-Execute architecture, end-to-end RAG, 5-metric eval framework, and open-source delivery for teams building agentic systems. 8+ years backend engineering at scale (1M+ records, 100K+ MAU, Python/AWS). Designed AgentProbe as a reusable toolkit - the same discipline as building shared frameworks for consistent delivery across customer engagements. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026). Demo: *github.com/venkatalakshmiparimala/AgentProbe*

CORE COMPETENCIES

LangGraph / LangChain | ReAct & Plan-Execute Agents | RAG Pipeline End-to-End | Eval Framework Design | LLM-as-Judge Pattern | Agent Debugging & Diagnosis | Shared Framework Development | Multi-Provider Frontier Models | Python / FastAPI / TypeScript | Observability & Tracing | Large-Scale Data Pipelines | AWS / Docker / CI/CD

EXPERIENCE

Entrata India (Db Xento Systems)

Jan 2017 - Aug 2024

Software Engineer → Associate Software Engineer → Trainee

Pune, India

- Built **production-grade software across 120+ Agile sprints** - strong practices in testing, CI/CD, documentation, and reliability for **100K+ monthly active users**; the enterprise-high bar this role requires.
- Built **20+ production modules and internal tools** (Python, TypeScript, PostgreSQL) prioritizing UX, reliability, and documentation - the same discipline as building reusable frameworks for consistent delivery across teams.
- Built **large-scale data processing frameworks** on AWS (S3, Glue, EC2) handling **1M+ records** with automated quality checks and zero security incidents - eliminating 80% of manual operational workload.
- Cut microservices execution time by **50%** via Docker containerization; maintained **99.5% uptime** for 6+ consecutive months via TDD/BDD quality gates.

George Mason University

Aug 2025 - May 2026

Graduate Teaching Assistant - Python & Databases

Fairfax, Virginia

- Mentored 30+ undergraduates through sprint-style feedback cycles; strong written and verbal communication translating technical concepts to non-technical audiences.

PROJECTS

AgentProbe: Production Agentic Workflow & Eval Framework OPEN SOURCE

Python · LangChain · LangGraph · FastAPI · ChromaDB · Ollama · React · Anthropic · OpenAI · Groq

Production agentic system built as a reusable framework for teams - not a prototype. Architecture: **LangGraph orchestration** implementing Plan-Execute (Orchestrator plans + dispatches) + ReAct recovery (Auto Fix reacts to failures, diagnoses, fixes); **RAG Tracer agent** enriches every run with per-stage attribution for agent behavior debugging; **5-metric eval framework** (correctness, hallucination, retrieval, completeness, compliance) measures success and regressions systematically - beyond trial-and-error; **Auto Fix agent** closes the loop: detects failure, attributes root cause, rewrites prompt, re-evaluates; **React observability dashboard**; **15-endpoint FastAPI backend**; multi-provider frontier model support (Anthropic, OpenAI, Groq, Ollama). Open-source, documented API - built for others to use. *github.com/venkatalakshmiparimala/AgentProbe*

AI on FHIR: NLP-Driven Medical Records Query System

Flask · LLMs · FHIR · React · Docker · spaCy · BERT · Embeddings · NumPy · Pandas

NLP pipeline mapping natural language queries (EN/ES) to structured FHIR APIs via LLMs, BERT, embeddings. HIPAA-compliant RBAC, containerized, structured output grounding reduces hallucination. **Reduced retrieval time ~60%**. *github.com/venkatalakshmiparimala/AI_on_FHIR*

CrashClarity: Predictive Modeling on 100K+ Records

Python · R · Random Forest · Logistic Regression · Feature Engineering · GitHub Pages

Full ML lifecycle on UK road accident data. **85%+ accuracy**; interactive dashboard for non-technical stakeholders.

EDUCATION

M.S. Data Analytics Engineering - George Mason University 2024-2026

GPA: 3.87 · Virginia, USA

B.Tech Computer Science & Engineering - Sree Venkateshwara College of Engineering 2013-2017

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, R, Bash, HTML/CSS

Agentic AI: LangChain, LangGraph, ReAct, Plan-Execute, RAG Pipelines, ChromaDB, LLM Evaluation, LLM-as-Judge, Eval Framework Design, Agent Debugging, Anthropic, OpenAI, Groq, Ollama, FastAPI, Prompt Engineering

ML Stack: PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, BERT, spaCy, Hugging Face, Feature Engineering

Data & Cloud: AWS (S3, EC2, RDS, Glue), Azure, PostgreSQL, MySQL, NoSQL, Docker, CI/CD (GitHub Actions), RabbitMQ

Testing & Quality: TDD/BDD, Integration Testing, Selenium, Git, JIRA, Postman, Agile/Scrum